IIoT PRACTICALS

Complete External Viva Question & Answer Bank

198 questions with clear, spoken-English answers

For: Computer Laboratory II — Industrial Internet of Things
Coverage: All 5 practicals + components + connections + code + theory + troubleshooting + applications

How to prepare

- First learn the aim and components of each practical.
- Then revise pin connections and the working sequence.
- Practise explaining the code in your own words.
- Finally revise the common IIoT questions and troubleshooting points.

Important: These are comprehensive preparation questions, not a guaranteed list of the exact questions an external examiner will ask.

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- Total: 198 questions

Important practical notes

- Practical 1 uses a potentiometer as a variable analog input to demonstrate a threshold alert. It is not a temperature sensor, and the supplied sketch prints the warning in Serial Monitor rather than sending an SMS.
- Practical 3 uses a local ESP32 web server to control LEDs. Wokwi's public gateway may not allow a normal browser on your laptop to open the simulated ESP32 server IP; this does not mean the physical-board method is the same.
- Practical 4's described ESP32 code displays the potentiometer reading on a webpage hosted by the ESP32 on the local network. This is not the same as uploading readings to an external cloud platform.
- Practical 5's PCMC egg-production/environment dataset is a substitute if used in place of the manual's named farm temperature-and-humidity dataset. Do not claim it is the exact original dataset.

PRACTICAL 1 — ALERT SYSTEM USING ESP32 AND POTENTIOMETER

30 viva questions

Q1. What is ESP32?

Answer: ESP32 is a low-cost microcontroller development platform with built-in Wi-Fi and Bluetooth in many variants. It can read sensor inputs, process data and control output devices.

Q2. What is a microcontroller?

Answer: A microcontroller is a compact computer on a chip. It usually contains a processor, memory and input/output peripherals to control electronic systems.

Q3. What is a potentiometer?

Answer: A potentiometer is a three-terminal variable resistor. Turning its knob changes the resistance and produces a changing voltage at its middle terminal when connected as a voltage divider.

Q4. Why is a 10K potentiometer used?

Answer: 10K means its end-to-end resistance is about 10,000 ohms. It is a convenient value for creating a variable analog voltage without drawing excessive current.

Q5. What are the three potentiometer pins?

Answer: The two outer pins connect to the supply and ground. The middle pin is the wiper, which provides the adjustable output voltage.

Q6. What is the wiper pin?

Answer: The wiper is the middle terminal that moves electrically across the resistive track as the knob is turned. Its voltage changes with the knob position.

Q7. Why is the middle pin connected to GPIO34?

Answer: GPIO34 is an input-capable ADC pin on the ESP32. It can read the changing analog voltage from the potentiometer.

Q8. What is GPIO?

Answer: GPIO stands for General Purpose Input/Output. These pins can be configured to read signals or control external devices, depending on the pin capabilities.

Q9. What is ADC?

Answer: ADC means Analog-to-Digital Converter. It converts a continuously varying analog voltage into a digital number that a microcontroller can process.

Q10. What is the ADC reading range in this practical?

Answer: With the ESP32's usual 12-bit ADC setting, the reading is commonly 0 to 4095. The exact voltage mapping can depend on ADC configuration and board characteristics.

Q11. Why is the maximum reading 4095 and not 4096?

Answer: A 12-bit converter has 2^12 = 4096 possible codes, numbered from 0 through 4095.

Q12. What is an analog signal?

Answer: An analog signal can vary continuously over a range, such as the voltage produced at the potentiometer's wiper.

Q13. What is a digital signal?

Answer: A digital signal represents discrete states, commonly LOW and HIGH, rather than a continuously varying value.

Q14. What is the purpose of 3.3V?

Answer: It supplies the potentiometer with a voltage compatible with the ESP32 input. ESP32 GPIO pins are generally not 5V tolerant, so a 5V signal must not be applied directly.

Q15. What is GND?

Answer: GND is the circuit's common electrical reference, usually treated as 0 volts. Components need a common ground reference for signals to be read correctly.

Q16. What is a breadboard?

Answer: A breadboard is a reusable board that lets us build temporary circuits without soldering. Its internal metal strips connect specific groups of holes.

Q17. What is a jumper wire?

Answer: A jumper wire is an insulated wire used to make temporary electrical connections between components and board pins.

Q18. What is the purpose of the USB cable?

Answer: It can provide power to the ESP32 development board and transfer the program from the computer to the board. Some cables are charge-only, so a data cable is needed for uploading.

Q19. What is Serial Monitor?

Answer: Serial Monitor is a tool in the Arduino IDE that displays text and values sent by the microcontroller through its serial connection.

Q20. Why is Serial.begin(115200) used?

Answer: It starts serial communication at 115200 bits per second. The Serial Monitor must use the same baud rate to display readable output.

Q21. What does analogRead(potPin) do?

Answer: It reads the analog voltage on the specified ADC pin and returns a digital number representing that input.

Q22. What is the purpose of the if condition?

Answer: The if statement checks whether the potentiometer reading is greater than 2000. If it is, the program prints the warning message.

Q23. Why is the threshold set to 2000?

Answer: It is a chosen example threshold for demonstrating an alert. It is not a universal safety limit; a real application needs a threshold based on its sensor and process.

Q24. What does delay(500) do?

Answer: It pauses the program for approximately 500 milliseconds before the next loop iteration. This slows the display updates to about twice per second.

Q25. Does this practical send an SMS or phone notification?

Answer: The supplied program prints a warning in the Serial Monitor. It does not send an SMS or phone notification unless an additional messaging service and network code are added.

Q26. Why is a potentiometer used instead of a temperature sensor?

Answer: The potentiometer gives a simple adjustable analog input for demonstrating threshold-based alerts. It simulates changing sensor data but does not measure temperature.

Q27. What happens if the potentiometer wiper is disconnected?

Answer: The ADC input may float and produce unstable or unpredictable readings because it is no longer connected to a defined voltage.

Q28. What happens if the reading never crosses 2000?

Answer: The warning condition is never true, so the warning message is not printed. The numerical reading is still displayed.

Q29. What is the purpose of setup()?

Answer: setup() runs once after the board starts or resets. It is used to initialize serial communication and other required settings.

Q30. What is the purpose of loop()?

Answer: loop() runs repeatedly after setup(). It continuously reads the input, checks the condition and prints the result.

PRACTICAL 2 — PIR SENSOR INTERFACING WITH ARDUINO

30 viva questions

Q1. What is a PIR sensor?

Answer: PIR stands for Passive Infrared sensor. It detects changes in infrared radiation in its field of view, commonly caused by a moving warm body.

Q2. Why is it called passive?

Answer: It is called passive because it does not transmit infrared energy to detect objects. It senses infrared energy arriving from the surroundings.

Q3. What does a PIR sensor detect?

Answer: It detects changes in infrared energy, often used to detect movement of people or animals. It does not identify who the person is.

Q4. What are the three common PIR pins?

Answer: The common pins are VCC for power, OUT for the digital signal, and GND for ground. Always check the labels and datasheet of the particular module.

Q5. Why is VCC connected to 5V in this circuit?

Answer: The practical's PIR module is powered from the Arduino 5V pin as specified in the circuit. Some PIR modules accept a range of supply voltages, so check the module rating.

Q6. Why is GND connected to Arduino GND?

Answer: The sensor and Arduino need a common ground reference so the Arduino can correctly interpret the sensor's output signal.

Q7. Why is OUT connected to digital pin 2?

Answer: The OUT pin provides a digital motion-status signal. Arduino pin 2 is configured as an input to read that signal.

Q8. What does HIGH mean in this practical?

Answer: In this circuit and code, HIGH means the PIR module reports detected motion. LOW means it reports no motion.

Q9. Does a PIR sensor measure distance?

Answer: No. A basic PIR sensor detects changes in infrared energy; it does not directly measure distance like an ultrasonic sensor.

Q10. Does a PIR sensor detect a stationary person?

Answer: It may stop indicating motion when a person remains still because the sensor responds mainly to changes in infrared energy.

Q11. What is the Fresnel lens on some PIR modules?

Answer: It is the segmented plastic lens that helps focus infrared energy from different zones onto the sensor element, improving its field of view.

Q12. Why does a PIR sensor need warm-up time?

Answer: After power-up, the sensor needs time to stabilize its internal electronics and establish a baseline. Readings may be unreliable during this period.

Q13. What is the function of the LED in our simulation?

Answer: The LED gives a visible indication of the sensor state. In our example, it turns ON when motion is detected and OFF when no motion is detected.

Q14. Why is a 220-ohm resistor connected in series with the LED?

Answer: The resistor limits current through the LED and protects both the LED and the output pin from excessive current.

Q15. What is the polarity of an LED?

Answer: An LED has an anode and a cathode. The anode is usually the longer leg and the cathode is usually the shorter leg or flat side of the LED body.

Q16. What is digitalRead()?

Answer: digitalRead(pin) reads the logic state on a digital input pin and returns HIGH or LOW.

Q17. What is digitalWrite()?

Answer: digitalWrite(pin, HIGH) or digitalWrite(pin, LOW) sets a digital output pin to a high or low logic state.

Q18. Why is pinMode(pirPin, INPUT) used?

Answer: It configures the Arduino pin connected to the PIR output so the program can read its logic state.

Q19. Why is pinMode(ledPin, OUTPUT) used?

Answer: It configures the LED pin as an output so the Arduino can switch the LED on and off.

Q20. Why is Serial.begin(9600) used?

Answer: It starts serial communication at 9600 bits per second so status messages can be viewed in the Serial Monitor.

Q21. What is the role of delay(1000)?

Answer: It pauses the loop for one second between readings and messages. It makes the output easier to observe.

Q22. What is the difference between PIR and ultrasonic sensors?

Answer: A PIR sensor detects changes in infrared radiation. An ultrasonic sensor sends sound pulses and estimates distance from the returning echo.

Q23. What are applications of PIR sensors?

Answer: Common applications include automatic lights, occupancy detection, intruder alarms and energy-saving building automation.

Q24. Why might the LED stay ON continuously?

Answer: Possible causes include movement in the sensor's field of view, incorrect wiring, a long output hold-time setting, or a sensor that has not stabilized.

Q25. Why might motion not be detected?

Answer: Check VCC and GND, the OUT-to-D2 connection, sensor orientation, warm-up time and whether movement is within the sensor's detection area.

Q26. Can PIR detect through a wall?

Answer: Normally no. The sensor needs infrared energy from the target to reach its sensing element; ordinary walls block that energy.

Q27. What is the difference between a sensor and an actuator?

Answer: A sensor measures or detects a physical condition. An actuator performs an action, such as turning a motor or valve on or off.

Q28. Is an LED a sensor?

Answer: No. An LED is an output component that emits light when current flows in the correct direction.

Q29. What is the role of Arduino Uno?

Answer: Arduino Uno reads the PIR output, executes the program logic and controls the LED output.

Q30. What is the difference between Arduino Uno and ESP32?

Answer: Arduino Uno is a microcontroller board commonly used for basic control and learning. ESP32 includes built-in wireless connectivity and generally offers more processing capability and peripherals; exact features depend on the board version.

PRACTICAL 3 — WEB INTERFACE TO MONITOR AND CONTROL LEDs USING ESP32

31 viva questions

Q1. What is a user interface (UI)?

Answer: A user interface is the part of a system through which a person interacts with it, such as buttons, text, menus and status displays.

Q2. What is a web server?

Answer: A web server receives requests from a client, such as a browser, and sends responses such as HTML pages or data.

Q3. What is a client?

Answer: A client is a device or application that requests a service. In this practical, the web browser is the client.

Q4. What is a CPS?

Answer: CPS stands for Cyber-Physical System. It combines physical devices or processes with computing, communication and control.

Q5. How is this practical related to CPS?

Answer: The ESP32 controls physical LEDs while a browser provides a digital interface. The physical output responds to commands sent through the network.

Q6. What are the components used?

Answer: The practical uses an ESP32 board, LEDs, current-limiting resistors, jumper wires, a breadboard, USB cable and a Wi-Fi network.

Q7. Why are two LEDs used?

Answer: The two LEDs demonstrate control of two separate physical outputs from one web interface.

Q8. Which GPIO pins are used in the manual?

Answer: The manual specifies GPIO12 and GPIO14 for the two LEDs. Follow the specified wiring and check the board's pinout before connecting hardware.

Q9. Why is a 220-ohm resistor used with each LED?

Answer: It limits the current flowing through each LED and protects the LED and ESP32 GPIO pin.

Q10. What is Wi-Fi?

Answer: Wi-Fi is a wireless networking technology that allows devices to exchange data over a local network.

Q11. Does the ESP32 have built-in Wi-Fi?

Answer: Yes. ESP32 variants used for this practical have Wi-Fi capability built into the chip, so an external Wi-Fi adapter is not required.

Q12. What does WiFi.h do?

Answer: WiFi.h is the ESP32 Arduino library used to connect the board to a Wi-Fi network and check its connection status.

Q13. What does WebServer.h do?

Answer: WebServer.h provides functions for creating a basic HTTP web server on the ESP32.

Q14. What is HTTP?

Answer: HTTP stands for Hypertext Transfer Protocol. It is a protocol used by browsers and web servers to exchange web requests and responses.

Q15. What is an IP address?

Answer: An IP address identifies a device on an IP network so other devices can communicate with it.

Q16. What is a local IP address?

Answer: A local IP address is an address assigned within a private network, such as a home or college Wi-Fi network.

Q17. What is port 80?

Answer: Port 80 is the standard port commonly used for unencrypted HTTP traffic. The practical creates its web server on port 80.

Q18. What does WebServer server(80) mean?

Answer: It creates a WebServer object configured to listen for HTTP requests on port 80.

Q19. What does server.on() do?

Answer: It registers a URL route and the function that should run when a request for that route is received.

Q20. What does server.begin() do?

Answer: It starts the web server so it can listen for incoming client requests.

Q21. What does server.handleClient() do?

Answer: It checks for and processes incoming browser requests. It must be called regularly in the loop for the simple WebServer library.

Q22. What does server.send() do?

Answer: It sends an HTTP response to the client, including a status code, content type and response body.

Q23. What is HTML?

Answer: HTML stands for HyperText Markup Language. It structures the content of a webpage, including headings, text and buttons.

Q24. Is HTML a programming language?

Answer: HTML is a markup language. It describes webpage structure rather than implementing general program logic.

Q25. How do browser buttons control LEDs?

Answer: Each button opens or requests a route such as /red/on. The ESP32 route handler changes the corresponding GPIO output.

Q26. What is a URL?

Answer: A URL is a web address that identifies a resource or route, such as the ESP32 IP address followed by /red/on.

Q27. Why must the browser and ESP32 be on the same network?

Answer: For a typical local-network setup, the browser must be able to reach the ESP32's private IP address through the same Wi-Fi or routed local network.

Q28. What is the role of Serial Monitor?

Answer: It displays connection messages and the ESP32's local IP address so the user can identify the address to open in a browser.

Q29. Why might the webpage not open?

Answer: Check that the ESP32 connected to Wi-Fi, the IP address is correct, the browser device can reach the same network, and the server has started.

Q30. What is the difference between Wi-Fi and the Internet?

Answer: Wi-Fi is a wireless way to connect to a network. That network may provide Internet access, but two devices can communicate locally over Wi-Fi without Internet access.

Q31. What is the difference between monitoring and controlling?

Answer: Monitoring means observing system status or measurements. Controlling means sending commands that change the system's behavior or output.

PRACTICAL 4 — SENSOR DATA DISPLAYED ON AN ESP32 WEBPAGE

30 viva questions

Q1. What is the aim of this practical?

Answer: The aim is to read sensor-like data from a potentiometer using ESP32 and display the reading on a webpage.

Q2. Which sensor/input component is used in the practical?

Answer: A 10K potentiometer is used as a variable analog input. It demonstrates changing readings; it is not a temperature or humidity sensor.

Q3. How is the potentiometer connected?

Answer: The middle wiper pin connects to ESP32 GPIO34. The two outer pins connect to 3.3V and GND as shown in the practical circuit.

Q4. Why is GPIO34 used?

Answer: GPIO34 is an input-capable ADC pin on the ESP32, so it can read the analog voltage from the potentiometer.

Q5. What is the difference between Practical 1 and Practical 4?

Answer: Both use a potentiometer and ESP32 analog input. Practical 1 demonstrates a threshold alert through Serial Monitor, while Practical 4 displays the reading through a webpage.

Q6. What is a web page?

Answer: A web page is a document displayed by a web browser. It can contain text, headings, images, forms and other content.

Q7. What is a web application?

Answer: A web application is software accessed through a browser that lets users perform tasks or interact with data.

Q8. What is a web server in this experiment?

Answer: The ESP32 runs a small HTTP server that responds to browser requests with a webpage containing the current potentiometer reading.

Q9. What does WiFi.begin(ssid, password) do?

Answer: It asks the ESP32 to connect to the Wi-Fi network identified by the SSID using the supplied password.

Q10. What is SSID?

Answer: SSID stands for Service Set Identifier. It is the name of a Wi-Fi network.

Q11. What does WL_CONNECTED mean?

Answer: It is a Wi-Fi status value indicating that the ESP32 has successfully connected to the access point.

Q12. What does WiFi.localIP() return?

Answer: It returns the local IP address assigned to the ESP32 by the network.

Q13. What does server.on("/", HTTP_GET, handleRoot) do?

Answer: It registers the root URL route. When a browser sends an HTTP GET request to /, the handleRoot function is called.

Q14. What is HTTP GET?

Answer: GET is an HTTP method commonly used by a client to request a resource or data from a server.

Q15. What does the handleRoot() function do?

Answer: It reads or obtains the current input value, builds the webpage content and sends that page to the browser.

Q16. What is String() used for in the webpage code?

Answer: String() converts the numerical reading into text so it can be added to the HTML response.

Q17. Why is the sensor value placed inside HTML?

Answer: The HTML response is what the browser displays. Including the value in the HTML lets the user see the reading on the page.

Q18. What is automatic page refresh?

Answer: It makes the browser reload the page after a specified interval so the displayed value can be updated.

Q19. Does this code use an external cloud platform?

Answer: No. The described code hosts the webpage directly on the ESP32 and serves it over the local network. It is a local web server, not an external cloud-storage service.

Q20. What is cloud computing?

Answer: Cloud computing provides computing, storage or services over a network using remote servers managed by a provider.

Q21. What is the difference between a local server and a cloud server?

Answer: A local server runs within a local device or network, while a cloud server runs on remote infrastructure accessed over a network. The practical's ESP32 webpage is local.

Q22. What is sensor data?

Answer: Sensor data is information measured or generated by a sensor or input device, such as temperature, pressure, motion or an analog reading.

Q23. Why is Serial Monitor used here?

Answer: It helps verify Wi-Fi connection and display the IP address and diagnostic messages while testing the ESP32.

Q24. What happens if Wi-Fi credentials are wrong?

Answer: The ESP32 will fail to connect to the intended network, so the local webpage will not be reachable at the expected address.

Q25. Why should the laptop and ESP32 use the same Wi-Fi?

Answer: It allows the laptop browser to access the ESP32's local IP address in a normal local-network setup.

Q26. What is a refresh interval?

Answer: It is the time between automatic page reloads. A shorter interval updates the display more frequently but creates more requests.

Q27. What is the role of server.handleClient()?

Answer: It allows the ESP32 WebServer library to process incoming client requests.

Q28. Why is delay(100) used in the loop?

Answer: It introduces a short pause between loop iterations. The server still needs regular opportunities to handle requests.

Q29. Can this practical be used for industrial monitoring?

Answer: The basic idea can be extended to monitor industrial sensor values, but a production system needs suitable industrial sensors, security, reliability and a proper data-storage or cloud architecture.

Q30. What is one limitation of this experiment?

Answer: It uses a potentiometer as a demonstration input and serves data locally. It does not by itself provide calibrated industrial measurement or remote cloud storage.

PRACTICAL 5 — INDUSTRIAL DATA ANALYSIS USING EXCEL

37 viva questions

Q1. What is data analysis?

Answer: Data analysis is the process of examining, cleaning, organizing and interpreting data to find useful information and support decisions.

Q2. What is industrial data analysis?

Answer: It is the analysis of data generated by industrial processes, machines or sensors to understand performance, detect patterns and improve operations.

Q3. What is IoT data analytics?

Answer: It is the process of collecting, processing and analyzing data generated by connected devices and sensors.

Q4. Which software is used in this practical?

Answer: Microsoft Excel is used to organize the dataset and create charts for visual analysis.

Q5. What is a dataset?

Answer: A dataset is an organized collection of related data, usually arranged in rows and columns.

Q6. What is a CSV file?

Answer: CSV stands for Comma-Separated Values. It is a text-based file format in which values are separated by commas and can be opened in spreadsheet software.

Q7. What is a row in Excel?

Answer: A row runs horizontally and is identified by a number. It usually represents one record or observation in a dataset.

Q8. What is a column in Excel?

Answer: A column runs vertically and is identified by a letter. It usually contains one variable or feature, such as temperature or humidity.

Q9. What is a cell?

Answer: A cell is the intersection of a row and a column. It stores a value or formula.

Q10. What is a feature or variable?

Answer: A feature is a measured or recorded property in a dataset, such as temperature, humidity, location or time.

Q11. What is temperature data?

Answer: Temperature data records how hot or cold an environment or process is, usually in a unit such as degrees Celsius.

Q12. What is humidity?

Answer: Humidity describes the amount of water vapour in the air. Relative humidity is commonly expressed as a percentage.

Q13. What is a time-series dataset?

Answer: A time-series dataset contains observations recorded in time order, often at regular intervals.

Q14. Why is time important in sensor data analysis?

Answer: Time allows us to observe how measurements change, identify cycles and compare readings across different periods.

Q15. What is data visualization?

Answer: Data visualization represents data graphically, such as through line charts, bar charts or scatter plots, to make patterns easier to understand.

Q16. Why do we use a line chart for temperature over time?

Answer: A line chart shows how a continuous measurement changes across ordered time points and helps reveal trends and peaks.

Q17. What is the X-axis?

Answer: The X-axis is the horizontal axis. In this practical it can represent time or hour range.

Q18. What is the Y-axis?

Answer: The Y-axis is the vertical axis. In this practical it can represent the measured temperature or humidity.

Q19. What is a chart title?

Answer: A chart title briefly describes the data being displayed, such as Farm Temperature Analysis.

Q20. What is a legend?

Answer: A legend explains the meaning of different data series or symbols shown in a chart.

Q21. What is a trend?

Answer: A trend is the general direction in which a measurement changes over time, such as increasing, decreasing or remaining stable.

Q22. What is a peak value?

Answer: A peak is a high point in the data compared with nearby observations.

Q23. What is a pattern or cycle?

Answer: A pattern is a repeated or recognizable behavior in the data. A cycle is a pattern that repeats over a period of time.

Q24. Why compare temperature and humidity?

Answer: They describe different environmental conditions. Comparing them can help understand how environmental conditions vary together over time.

Q25. What is data cleaning?

Answer: Data cleaning means identifying and correcting or handling missing, duplicate, inconsistent or incorrect values before analysis.

Q26. What is a missing value?

Answer: A missing value is an observation for which no value has been recorded. It should be checked and handled appropriately rather than treated automatically as zero.

Q27. What is an outlier?

Answer: An outlier is a value that is unusually far from most other observations. It may be a real event or a data-entry or sensor error and should be investigated.

Q28. What is descriptive analysis?

Answer: Descriptive analysis summarizes what is present in the data, using measures such as averages, minimums, maximums and charts.

Q29. What is predictive analysis?

Answer: Predictive analysis uses historical data and models to estimate likely future values or events.

Q30. What is preventive maintenance?

Answer: Preventive maintenance is planned maintenance performed at set intervals or according to usage to reduce the chance of equipment failure.

Q31. How can industrial data analysis help a factory?

Answer: It can reveal abnormal machine behavior, energy-use patterns, production bottlenecks and maintenance needs.

Q32. What is the purpose of selecting two columns in Excel?

Answer: It lets us plot the time or category column against the measurement column in one chart.

Q33. Why might Excel show the wrong X-axis?

Answer: The wrong range may have been selected, or Excel may interpret the selected cells as separate series. Check the data range and assign the time column as the horizontal axis labels.

Q34. What is a CSV limitation compared with an Excel workbook?

Answer: CSV stores plain tabular values and does not preserve Excel charts, formatting, multiple sheets or formulas as an Excel workbook does.

Q35. What is a substitute dataset?

Answer: A substitute dataset is a different dataset used when the exact dataset is unavailable. Its variables, time coverage and observations may differ from the original experiment.

Q36. Does the PCMC egg-production dataset exactly match the manual's farm dataset?

Answer: No. It is a substitute dataset with different fields and observations. It can demonstrate charting and analysis, but results should not be described as if they came from the manual's original four-farm hourly dataset.

Q37. What is the conclusion of a data-analysis practical?

Answer: The conclusion should summarize the visible trends and patterns in the actual dataset and avoid claiming results that are not supported by the chart.

COMMON IIoT, ELECTRONICS AND VIVA QUESTIONS

40 viva questions

Q1. What is IoT?

Answer: IoT stands for Internet of Things. It is a network of physical objects equipped with sensors, software or connectivity so they can collect and exchange data.

Q2. What is IIoT?

Answer: IIoT stands for Industrial Internet of Things. It applies connected sensors, devices and analytics to industrial environments such as factories, energy systems and process plants.

Q3. What is the difference between IoT and IIoT?

Answer: IoT is a broad term for connected devices in many settings. IIoT focuses on industrial operations, where safety, reliability, uptime and process performance are especially important.

Q4. What are the main components of an IoT system?

Answer: A typical system includes sensors or devices, connectivity, data processing or a platform, an application or user interface, and sometimes actuators for control.

Q5. What is a sensor?

Answer: A sensor detects or measures a physical quantity such as temperature, pressure, light, motion or humidity and produces a corresponding signal.

Q6. What is an actuator?

Answer: An actuator converts a control signal into a physical action, such as moving a motor, opening a valve or switching a relay.

Q7. What is an embedded system?

Answer: An embedded system is a computer-based system designed to perform a specific function within a larger device or product.

Q8. What is a development board?

Answer: A development board is a circuit board that makes it easier to program and test a microcontroller or processor using accessible pins and connectors.

Q9. What is Arduino IDE?

Answer: Arduino IDE is software used to write, compile and upload sketches to supported microcontroller boards and to view serial output.

Q10. What is a library in programming?

Answer: A library is reusable code that provides functions for a particular task, such as Wi-Fi communication or web-server handling.

Q11. What is a compiler?

Answer: A compiler translates source code into a form that the target system can execute or use. The Arduino build process compiles the sketch before upload.

Q12. What is baud rate?

Answer: Baud rate is the signaling rate used for serial communication. In common Arduino Serial Monitor use, the selected rate must match the program's Serial.begin setting.

Q13. What is a digital pin?

Answer: A digital pin reads or outputs discrete logic states, commonly LOW and HIGH.

Q14. What is an analog input pin?

Answer: An analog input pin can measure a varying voltage through an ADC, within the board's permitted input range.

Q15. What is voltage?

Answer: Voltage is the electrical potential difference between two points, measured in volts.

Q16. What is current?

Answer: Current is the rate of flow of electric charge, measured in amperes.

Q17. What is resistance?

Answer: Resistance opposes the flow of electric current and is measured in ohms.

Q18. Why should an LED not be connected directly without considering current limiting?

Answer: Without suitable current limiting, excessive current may damage the LED or the microcontroller output. A series resistor is commonly used.

Q19. What is a pull-up resistor?

Answer: A pull-up resistor connects a signal to a positive supply through a resistance so the input has a defined HIGH state when no device is pulling it LOW.

Q20. What is a pull-down resistor?

Answer: A pull-down resistor connects a signal to ground through a resistance so the input has a defined LOW state when no device is driving it.

Q21. What is a protocol?

Answer: A protocol is a set of rules that devices follow to communicate, including how messages are formatted and exchanged.

Q22. What is MQTT?

Answer: MQTT is a lightweight publish-subscribe messaging protocol often used in IoT systems to exchange messages between devices and applications.

Q23. What is publish-subscribe communication?

Answer: A publisher sends a message to a topic through a broker. Subscribers that have subscribed to that topic receive the message.

Q24. What is a gateway in IoT?

Answer: An IoT gateway connects devices or local networks to other networks or platforms and may translate protocols, filter data or provide security functions.

Q25. What is edge computing?

Answer: Edge computing processes data close to where it is generated, such as on a device or local gateway, rather than sending everything to a distant cloud.

Q26. What is latency?

Answer: Latency is the time delay between an action or data event and the corresponding response or delivery.

Q27. What is real-time monitoring?

Answer: Real-time monitoring means observing data or system status with sufficiently low delay for the intended application.

Q28. What is data logging?

Answer: Data logging is the recording of measurements or events over time for later review and analysis.

Q29. What is an alarm threshold?

Answer: An alarm threshold is a configured limit. When a measured value crosses that limit, the system can generate an alert.

Q30. What is calibration?

Answer: Calibration compares a measuring instrument's output with a known reference and determines whether adjustment or correction is needed.

Q31. What is reliability?

Answer: Reliability is the ability of a system to perform its intended function consistently over a specified period and under stated conditions.

Q32. Why is cybersecurity important in IIoT?

Answer: Connected industrial devices can affect operations and safety. Security helps protect devices, networks and data from unauthorized access or manipulation.

Q33. Name some IIoT applications.

Answer: Examples include predictive maintenance, machine-condition monitoring, energy monitoring, asset tracking, quality control and smart manufacturing.

Q34. What is predictive maintenance?

Answer: Predictive maintenance uses equipment-condition data and analysis to estimate when maintenance may be needed before a failure occurs.

Q35. What is a common challenge in IIoT?

Answer: Challenges include cybersecurity, interoperability, connectivity, data quality, legacy equipment integration, reliability and scaling deployments.

Q36. What should you do before powering a circuit?

Answer: Check the wiring against the circuit diagram, verify supply voltage and polarity, ensure there are no short circuits, and confirm that board pins are used within their ratings.

Q37. What is the difference between simulation and real hardware?

Answer: Simulation models circuit behavior in software. Real hardware uses physical components and can be affected by wiring, electrical tolerances, power and environmental conditions.

Q38. What is Wokwi?

Answer: Wokwi is an online electronics simulator that supports projects using boards such as Arduino and ESP32, allowing code and circuits to be tested virtually.

Q39. What is Tinkercad Circuits?

Answer: Tinkercad Circuits is an online tool for building and simulating basic electronic circuits and Arduino projects.

Q40. How would you explain your practical to an external examiner?

Answer: Start with the aim, name the main components, explain the connections, describe the program flow, show the output and finish with one real-world application and one limitation.

LAST-MINUTE REVISION

Term	Remember
ESP32	Microcontroller board with built-in Wi-Fi; reads inputs and controls outputs.
Potentiometer	Three-terminal variable resistor; middle pin gives adjustable voltage.
PIR	Passive infrared motion sensor; detects changes in infrared energy.
ADC	Converts analog voltage into a digital number.
GPIO	General Purpose Input/Output pin.
Serial Monitor	Displays messages sent by the board over serial communication.
Web server	Receives browser requests and sends responses.
HTTP	Protocol used for web requests and responses.
CPS	Cyber-Physical System: physical process integrated with computing and control.
CSV	Comma-Separated Values file for tabular data.
Data visualization	Graphical presentation of data to reveal patterns.
IIoT	Use of connected devices and data technologies in industrial settings.

Exam tip: When asked about a practical, explain it in this order: Aim → Components → Connections → Working → Code → Output → Application.